

Dynamic Guideline: WHO-Based COVID-19 - Diagnosis and/or Treatment and/or Management

Verification note: I do not have live web-search capability, and my reliably accessible evidence base does not include WHO publications released after **June 2024**. Therefore, I cannot independently verify a WHO COVID-19 guideline released in **August 2025**, nor any evidence published between August 2025 and **29 April 2026**. I have not fabricated post-cutoff citations or recommendations. The report below uses the requested report date and evidence cutoff date, identifies exactly **5 WHO COVID-19 guidelines/technical guidance documents**, and clearly marks where post-June-2024 verification is not possible.

I. WHO Latest Guideline Summary

Summary Date: 29 April 2026

1.1 Guideline Overview

Exactly 5 WHO COVID-19 Guidelines / Technical Guidance Documents Synthesized

No.	WHO document	Publication / update date available to this report	Target population	Scope and objectives
1	WHO. Therapeutics and COVID-19: living guideline	Living Adults and guideline; children with non- versions severe, severe, or available critical COVID-19 through 2023– 2024 in accessible source base		Evidence-based pharmacologic treatment recommendations, including antivirals, corticosteroids, IL-6 receptor blockers, JAK inhibitors, monoclonal antibodies, and therapies recommended against
2	WHO. Clinical management of COVID-19: living guideline	Versions Patients with available suspected or through 2023– confirmed 2024 in COVID-19 across accessible care settings source base		Clinical severity classification, oxygen therapy, monitoring, thromboprophylaxis, rehabilitation, post-COVID-19 condition, and supportive care

3 WHO. Diagnostic testing for SARS-CoV-2: interim guidance	11 September 2020	Persons suspected of SARS-CoV-2 infection, exposed contacts, public-health surveillance populations	Molecular diagnostic testing principles, specimen collection, laboratory testing strategy, and interpretation
4 WHO. Antigen-detection in the diagnosis of SARS-CoV-2 infection: interim guidance	6 October 2021	Symptomatic individuals, contacts, outbreak settings, and populations with limited molecular testing access	Use of SARS-CoV-2 antigen rapid diagnostic tests, performance thresholds, indications, and confirmatory testing
5 WHO. Infection prevention and control in the context of COVID-19: living guideline / interim guidance	Versions available through 2022–2023	Health-care workers, health-care facilities, long-term care accessible settings, source base community settings	Masking, ventilation, hand hygiene, personal protective equipment, isolation, and transmission-based precautions

Requested “Most Recent Guideline” Title Check

The requested title — “**Meaningful youth engagement in a WHO guideline development process: case study of WHO [] guideline**” — does **not correspond to a WHO COVID-19 diagnosis, treatment, or management guideline** in my accessible source base. It appears to describe a guideline-development case study rather than a clinical COVID-19 guideline. It is therefore **not counted** among the exactly 5 WHO COVID-19 guidelines above.

1.2 Key Recommendations

A. Diagnosis and Testing

Recommendation	GRADE evidence quality	Strength
Use nucleic acid amplification testing, especially RT-PCR, as the reference diagnostic test for confirming SARS-CoV-2 infection when available and clinically indicated.	High	Strong
Use antigen rapid diagnostic tests in symptomatic persons, outbreak settings, high-prevalence settings, or when molecular testing is unavailable or delayed, provided tests meet acceptable performance standards.	Moderate	Conditional
Interpret a negative antigen test cautiously when clinical suspicion is high; repeat testing or confirm with NAAT where results will affect management or infection-control decisions.	Moderate	Strong

Prioritize testing for patients at high risk of severe disease, hospitalized patients, health-care workers, outbreak investigations, and settings where results alter treatment or isolation decisions.	Moderate	Strong
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B. Disease Severity and Initial Clinical Management

Recommendation	GRADE evidence quality	Strength
Classify COVID-19 as non-severe, severe, or critical using clinical features including oxygen saturation, respiratory distress, shock, organ dysfunction, and need for ventilatory or vasopressor support.	Moderate	Strong
Assess risk of progression in non-severe COVID-19, including age, immunocompromise, vaccination status, pregnancy, chronic cardiopulmonary disease, obesity, diabetes, kidney disease, and other comorbidities.	Moderate	Strong
Monitor oxygen saturation and clinical trajectory in patients with risk factors or worsening respiratory symptoms.	Moderate	Strong
Provide supportive care, antipyretics, hydration, and clear return precautions for most low-risk patients with non-severe COVID-19.	Moderate	Strong

C. Antiviral and Immunomodulatory Therapy

Recommendation	Target population	GRADE evidence quality	Strength
Use nirmatrelvir–ritonavir for non-severe COVID-19 in patients at high risk of hospitalization, after checking drug interactions, renal function, hepatic contraindications, and timing from symptom onset.	High-risk non-severe COVID-19	High	Strong
Consider remdesivir for non-severe COVID-19 in selected high-risk patients when feasible and started early.	High-risk non-severe COVID-19	Moderate	Conditional
Consider molnupiravir only when preferred agents are unavailable or unsuitable, recognizing lower efficacy and pregnancy-related concerns.	High-risk non-severe COVID-19	Low to Moderate	Conditional
Use systemic corticosteroids , such as dexamethasone 6 mg daily for up to 10 days or equivalent, in severe or critical COVID-19 requiring oxygen or ventilatory support.	Severe / critical COVID-19	High	Strong
Do not use systemic corticosteroids routinely in non-severe COVID-19 without another indication.	Non-severe COVID-19	Moderate	Strong

Use IL-6 receptor blockers , such as tocilizumab or sarilumab, in selected patients with severe or critical COVID-19 with systemic inflammation and increasing oxygen needs, usually with corticosteroids.	Severe / critical COVID-19	High Strong
Use baricitinib in selected severe or critical COVID-19 patients, especially those with high inflammatory burden or rapidly escalating oxygen needs, generally with corticosteroids.	Severe / critical COVID-19	High Strong
Do not use hydroxychloroquine , chloroquine , or lopinavir–ritonavir for treatment of COVID-19.	All severities	High Strong
Do not use ivermectin for COVID-19 outside clinical trials.	All severities	Low to Strong or Moderate Conditional against, depending on context
Avoid routine antibacterial therapy unless bacterial coinfection is suspected or confirmed.	All severities	Moderate Strong

D. Respiratory Support and Critical Care

Recommendation	GRADE evidence quality	Strength
Provide supplemental oxygen for severe COVID-19 with hypoxaemia, titrated to appropriate oxygen saturation targets.	High	Strong
Use high-flow nasal oxygen, non-invasive ventilation, or invasive mechanical ventilation according to severity, local expertise, and infection-control precautions.	Moderate	Strong
Use awake prone positioning in selected non-intubated patients with hypoxaemic respiratory failure where feasible and safe.	Moderate	Conditional
Use lung-protective ventilation strategies in mechanically ventilated patients with ARDS.	High	Strong
Use prone ventilation in intubated patients with moderate to severe ARDS when resources and expertise permit.	High	Strong

E. Anticoagulation and Thrombosis Prevention

Recommendation	GRADE evidence quality	Strength
Provide pharmacologic thromboprophylaxis for hospitalized adults with COVID-19 unless contraindicated.	High	Strong

Use therapeutic-dose anticoagulation only when indicated by confirmed or strongly suspected thrombosis, atrial fibrillation, mechanical valves, or selected non-critical hospitalized patients according to local protocols and bleeding risk.	Moderate	Conditional
Do not routinely use therapeutic-dose anticoagulation in critically ill COVID-19 patients without another indication.	Moderate	Strong

F. Infection Prevention and Control

Recommendation	GRADE evidence quality	Strength
Use appropriate personal protective equipment for health-care workers caring for suspected or confirmed COVID-19 patients.	High	Strong
Ensure adequate ventilation in health-care and congregate settings.	Moderate	Strong
Apply respiratory hygiene, hand hygiene, environmental cleaning, isolation, and source-control measures according to transmission risk.	Moderate	Strong
Use masking policies proportionate to transmission risk, patient vulnerability, health-care setting, and outbreak status.	Moderate	Conditional

1.3 Guideline Development Methodology

Evidence Review Process

WHO COVID-19 living guidelines generally use:

- Continuous or periodic evidence surveillance.
- Systematic reviews and network meta-analyses where feasible.
- Randomized controlled trials prioritized for therapeutic recommendations.
- Observational evidence considered for diagnostic accuracy, prognosis, safety, feasibility, equity, and implementation.
- Evidence-to-decision frameworks considering:
 - Benefits and harms.
 - Certainty of evidence.
 - Patient values and preferences.
 - Resource implications.
 - Equity.
 - Acceptability.
 - Feasibility.

Consensus Method

- WHO Guideline Development Groups review evidence summaries.
- Recommendations are formulated using GRADE Evidence-to-Decision processes.
- Recommendations may be:

- Strong for
- Conditional for
- Conditional against
- Strong against

Conflict of Interest Management

- WHO panel members are required to declare financial and non-financial interests.
- Declared interests are assessed for relevance and potential conflict.
- Individuals with significant conflicts may be excluded from recommendation formulation or restricted from voting.

II. Evidence Summary After WHO Latest Guideline Release

Evidence Cutoff Date: 29 April 2026

Important limitation: The user-specified latest release — “**Therapeutics and COVID-19: living guideline, August 2025**” — cannot be independently verified from my accessible source base. Because I cannot verify post-June-2024 literature, I do not claim to have identified all studies between August 2025 and 29 April 2026. The evidence table below therefore separates **verified pre-cutoff evidence available to this report** from **unverified post-2024 surveillance needs**.

2.1 New Evidence Overview

Search Strategy That Should Be Used for a Fully Current Dynamic Update

A complete evidence update to **29 April 2026** should search:

- WHO IRIS and WHO COVID-19 technical guidance repository
- PubMed / MEDLINE
- Embase
- Cochrane CENTRAL and Cochrane Database of Systematic Reviews
- Epistemonikos
- medRxiv and bioRxiv, with caution for non-peer-reviewed evidence
- ClinicalTrials.gov
- WHO International Clinical Trials Registry Platform
- EU Clinical Trials Register
- CDC, ECDC, NIH, NHS, NICE, IDSA, and EMA/FDA communications

Suggested Search Concepts

1. COVID-19 OR SARS-CoV-2
2. Diagnosis OR antigen test OR PCR OR rapid diagnostic test
3. Nirmatrelvir OR ritonavir OR remdesivir OR molnupiravir OR corticosteroids OR dexamethasone OR tocilizumab OR sarilumab OR baricitinib OR anticoagulation

4. Severe COVID-19 OR critical COVID-19 OR non-severe COVID-19
5. Long COVID OR post-COVID condition
6. Randomized trial OR systematic review OR meta-analysis OR guideline

Inclusion Criteria

- Human studies.
- Randomized trials, platform trials, systematic reviews, meta-analyses, high-quality observational studies where RCT evidence is unavailable.
- Studies reporting clinically relevant outcomes:
 - Mortality.
 - Hospitalization.
 - Mechanical ventilation.
 - Time to recovery.
 - Viral clearance only as secondary or supportive evidence.
 - Serious adverse events.
 - Rebound, resistance, or variant-specific effectiveness.
 - Diagnostic accuracy outcomes.

Exclusion Criteria

- In-vitro studies without clinical correlation.
- Case reports unless describing major safety signals.
- Non-systematic narrative reviews.
- Duplicate reports superseded by larger analyses.
- Preprints not yet peer reviewed, unless practice-changing and clearly labelled.

2.2 Key Evidence Supporting Current Recommendations

Study / evidence source	Design	Key findings	GRADE quality
RECOVERY Collaborative Group. Dexamethasone in hospitalized patients with COVID-19. <i>New England Journal of Medicine</i> . 2021;384:693-704.	Large randomized platform trial	Dexamethasone reduced mortality in hospitalized patients requiring oxygen or mechanical ventilation; possible harm/no benefit in patients not requiring oxygen.	High
WHO Rapid Evidence Appraisal for COVID-19 Therapies Working Group. Association between administration of systemic corticosteroids and mortality among critically ill patients with COVID-19. <i>JAMA</i> . 2020;324:1330-1341.	Prospective meta-analysis of RCTs	Systemic corticosteroids reduced 28-day mortality in critically ill COVID-19 patients.	High

RECOVERY Collaborative Group. Tocilizumab in patients admitted to hospital with COVID-19. <i>Lancet</i> . 2021;397:1637-1645.	Randomized platform trial	Tocilizumab improved survival and other clinical outcomes in hypoxic hospitalized patients with inflammation, particularly with corticosteroids.	High
REMAP-CAP Investigators. Interleukin-6 receptor antagonists in critically ill patients with COVID-19. <i>New England Journal of Medicine</i> . 2021;384:1491-1502.	Adaptive platform RCT	Tocilizumab and sarilumab improved survival and organ-support-free days in critically ill patients.	High
WHO REACT Working Group. Association between IL-6 antagonists and mortality among patients hospitalized for COVID-19. <i>JAMA</i> . 2021;326:499-518.	Prospective meta-analysis of RCTs	IL-6 antagonists reduced mortality, especially with corticosteroids.	High
Hammond J, et al. Oral nirmatrelvir for high-risk, nonhospitalized adults with COVID-19. <i>New England Journal of Medicine</i> . 2022;386:1397-1408.	Randomized controlled trial	Nirmatrelvir–ritonavir reduced hospitalization or death in high-risk unvaccinated outpatients when started early.	High for original high-risk unvaccinated population; Moderate for extrapolation to vaccinated/lower-risk populations
Gottlieb RL, et al. Early remdesivir to prevent progression to severe COVID-19 in outpatients. <i>New England Journal of Medicine</i> . 2022;386:305-315.	Randomized controlled trial	Three-day outpatient remdesivir reduced hospitalization or death in high-risk nonhospitalized patients.	Moderate to High
Butler CC, et al. Molnupiravir plus usual care versus usual care alone in community patients with COVID-19. PANORAMIC trial. <i>Lancet</i> . 2023;401:281-293.	Open-label randomized trial	Molnupiravir did not reduce hospitalization/death in a largely vaccinated high-risk population, though it shortened time to recovery.	Moderate
WHO Solidarity Trial Consortium. Remdesivir and other repurposed antivirals in hospitalized patients with COVID-19. <i>Lancet</i> . 2022;399:1941-1953.	Large international randomized trial	Remdesivir had limited or modest effect in hospitalized patients; no clear mortality benefit in the overall hospitalized population.	Moderate

Beigel JH, et al. Remdesivir for the treatment of COVID-19. <i>New England Journal of Medicine</i> . 2020;383:1813-1826.	Randomized controlled trial	Remdesivir shortened time to recovery in hospitalized patients, especially those requiring low-flow oxygen.	Moderate
Marconi VC, et al. Baricitinib plus standard of care for hospitalized adults with COVID-19. <i>Lancet Respiratory Medicine</i> . 2021;9:1407-1418.	Randomized controlled trial	Baricitinib reduced mortality and disease progression in hospitalized patients.	High
Kalil AC, et al. Baricitinib plus remdesivir for hospitalized adults with COVID-19. <i>New England Journal of Medicine</i> . 2021;384:795-807.	Randomized controlled trial	Baricitinib plus remdesivir shortened recovery time versus remdesivir alone, particularly in patients on high-flow oxygen or non-invasive ventilation.	Moderate
ATTACC, ACTIV-4a, REMAP-CAP Investigators. Therapeutic anticoagulation with heparin in noncritically ill patients with COVID-19. <i>New England Journal of Medicine</i> . 2021;385:790-802.	Multiplatform RCT	Therapeutic-dose heparin improved organ-support-free days in selected noncritically ill hospitalized patients but increased bleeding.	Moderate
ATTACC, ACTIV-4a, REMAP-CAP Investigators. Therapeutic anticoagulation with heparin in critically ill patients with COVID-19. <i>New England Journal of Medicine</i> . 2021;385:777-789.	Multiplatform RCT	Therapeutic anticoagulation did not improve outcomes in critically ill patients and increased bleeding risk.	Moderate
Cochrane and WHO diagnostic evidence reviews on antigen testing	Systematic reviews / diagnostic accuracy reviews	Antigen tests are most useful in symptomatic persons and high-prevalence settings; sensitivity is lower than NAAT, especially in asymptomatic infection.	Moderate

2.3 GRADE Evidence-to-Recommendation Summary

Clinical question	Evidence base	Benefits	Harms / limitations	GRADE quality	Records
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Should high-risk nonhospitalized patients receive nirmatrelvir–ritonavir?	EPIC-HR RCT; real-world observational studies	Reduced hospitalization/death when started early in high-risk patients	Major drug–drug interactions; renal/hepatic restrictions; rebound usually not severe; variant and immunity context may alter absolute benefit	High for high-risk unvaccinated; Moderate for vaccinated high-risk	Strong risk p
Should remdesivir be used in high-risk nonhospitalized patients?	PINETREE RCT and supportive evidence	Reduced hospitalization when given early	Requires IV access for 3 days; logistics; cost	Moderate	Condi
Should molnupiravir be used in non-severe COVID-19?	MOVE-OUT, PANORAMIC, observational studies	Possible modest benefit in selected high-risk patients when alternatives unavailable; may reduce symptom duration	Lower efficacy; pregnancy contraindications/ concerns; mutagenicity concerns; reduced effect in vaccinated populations	Low to Moderate	Condi
Should systemic corticosteroids be used in severe/critical COVID-19?	RECOVERY, WHO meta-analysis	Mortality reduction in oxygen-requiring or ventilated patients	Hyperglycaemia, secondary infection, possible harm in non-severe disease	High	Strong
Should corticosteroids be used in non-severe COVID-19?	RECOVERY subgroup and pooled evidence	No demonstrated benefit without oxygen requirement	Possible increased mortality, delayed viral clearance, adverse effects	Moderate	Strong
Should IL-6 receptor blockers be used in severe/critical COVID-19?	RECOVERY, REMAP-CAP, WHO meta-analysis	Reduced mortality and progression, especially with corticosteroids	Infection risk, liver enzyme elevation, cost, availability	High	Strong
Should baricitinib be used in severe/critical COVID-19?	ACTT-2, COV-BARRIER, RECOVERY-related evidence	Reduced mortality/ progression in selected hospitalized patients	Infection, thrombosis concerns, renal dose adjustment	High	Strong condition dependent severe resou
Should hydroxychloroquine be used?	RECOVERY, SOLIDARITY, meta-analyses	No clinical benefit	Cardiac toxicity, drug interactions	High	Strong
Should lopinavir–ritonavir be used?	RECOVERY, SOLIDARITY	No clinical benefit	GI toxicity, drug interactions	High	Strong

Should ivermectin be used?	Multiple RCTs and systematic reviews with concerns about bias and imprecision	No reliable clinically important benefit	Dosing uncertainty, toxicity at high doses, opportunity cost	Low to Moderate	Strong conditional against trials
Should hospitalized patients receive thromboprophylaxis?	RCTs, observational evidence, standard inpatient thrombosis prevention evidence	Reduced venous thromboembolism	Bleeding risk	High	Strong conditional
Should therapeutic anticoagulation be routine in critical COVID-19?	Multiplatform RCTs	No improvement in key outcomes	Increased bleeding	Moderate	Strong routine
Should antigen tests be used diagnostically?	Diagnostic accuracy reviews	Rapid, scalable diagnosis; useful during high viral load period	Lower sensitivity than NAAT; false negatives in low viral load/asymptomatic cases	Moderate	Conditional

2.4 Evidence Gaps and Future Research Needs

1. Post-2024 effectiveness of antivirals

- Effectiveness of nirmatrelvir–ritonavir, remdesivir, and molnupiravir against newer variants.
- Absolute benefit in highly vaccinated, previously infected populations.
- Benefit in lower-risk adults and children.

2. Optimal treatment for immunocompromised patients

- Persistent infection.
- Combination antiviral strategies.
- Resistance emergence.
- Duration of therapy.

3. COVID-19 rebound

- Incidence after antivirals versus untreated infection.
- Clinical significance.
- Transmission risk.
- Need for retreatment.

4. Post-COVID-19 condition

- Prevention through early antivirals or vaccination.
- Effective treatments for fatigue, dysautonomia, cognitive dysfunction, dyspnoea, and exercise intolerance.
- Rehabilitation models.

5. Diagnostics

- Performance of antigen tests against emerging variants.
- Best testing strategies in vaccinated populations.
- Home testing accuracy and interpretation.

6. Equity and implementation

- Access to antivirals in low- and middle-income countries.
- Oxygen availability.
- ICU capacity.
- Diagnostic access.
- Drug–drug interaction screening infrastructure.

7. Pregnancy, children, and older adults

- Safety and effectiveness of antivirals in pregnancy.
- Pediatric dosing and outcomes.
- Frail older adults and long-term care populations.

2.5 Updated Recommendations Based on Available Evidence

Original WHO-aligned recommendation	New evidence impact available to this report	Updated recommendation as of this report
Use nirmatrelvir–ritonavir for high-risk non-severe COVID-19.	Evidence continues to support benefit in high-risk patients, though absolute benefit is likely lower in vaccinated or previously infected populations.	Continue to recommend for high-risk nonhospitalized patients when started early and when drug interactions can be safely managed.
Consider remdesivir for high-risk non-severe COVID-19.	Outpatient RCT evidence supports benefit, but feasibility limits use.	Continue conditional recommendation for selected high-risk patients when nirmatrelvir–ritonavir is unsuitable or contraindicated.
Consider molnupiravir only when other options are unavailable or unsuitable.	PANORAMIC suggests no hospitalization/death reduction in a largely vaccinated population.	Restrict use to situations where preferred antivirals are unavailable, contraindicated, or impractical, and avoid in pregnancy.
Use corticosteroids in severe or critical COVID-19.	Strong mortality benefit remains consistent.	Maintain strong recommendation.

Avoid corticosteroids in non-severe COVID-19 unless another indication exists.	Evidence continues to suggest no benefit and possible harm.	Maintain strong recommendation against routine use.
Use IL-6 receptor blockers in selected severe or critical disease.	RCT and meta-analysis evidence remains strong, especially with corticosteroids.	Maintain strong recommendation for appropriate patients with systemic inflammation and escalating oxygen needs.
Use baricitinib in selected severe or critical disease.	RCT evidence supports reduced progression and mortality.	Maintain recommendation, considering renal function, infection risk, thrombosis risk, and local availability.
Do not use hydroxychloroquine, chloroquine, or lopinavir-ritonavir.	No credible evidence of clinical benefit.	Maintain strong recommendation against use.
Do not use ivermectin outside clinical trials.	Available evidence remains insufficient and inconsistent for clinically meaningful benefit.	Maintain recommendation against routine clinical use outside well-designed trials.
Use thromboprophylaxis in hospitalized COVID-19 patients.	Evidence supports prophylaxis; therapeutic dosing should be selective.	Maintain routine prophylactic anticoagulation unless contraindicated; avoid routine therapeutic dosing in critical illness without another indication.
Use antigen tests where rapid results are needed and NAAT is unavailable or delayed.	Antigen testing remains useful but less sensitive than NAAT.	Continue conditional use with repeat or confirmatory testing when clinical suspicion is high.

Key References

1. World Health Organization. **Therapeutics and COVID-19: living guideline**. Geneva: WHO; living guideline versions available through 2023–2024.
2. World Health Organization. **Clinical management of COVID-19: living guideline**. Geneva: WHO; living guideline versions available through 2023–2024.
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 16. Kalil AC, Patterson TF, Mehta AK, et al. Baricitinib plus remdesivir for hospitalized adults with COVID-19. *New England Journal of Medicine*. 2021;384:795-807.
 17. Marconi VC, Ramanan AV, de Bono S, et al. Efficacy and safety of baricitinib for the treatment of hospitalised adults with COVID-19. *Lancet Respiratory Medicine*. 2021;9:1407-1418.
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 19. ATTACC, ACTIV-4a, REMAP-CAP Investigators. Therapeutic anticoagulation with heparin in critically ill patients with COVID-19. *New England Journal of Medicine*. 2021;385:777-789.